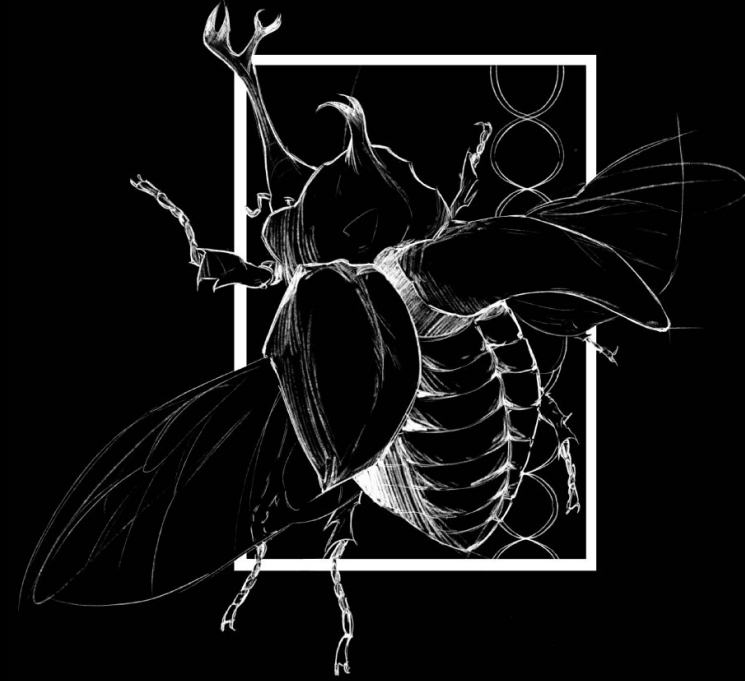




Background

Chromosome number and structure evolve in ways that profoundly shape genome organization, reproductive barriers, and the diversification of lineages across the tree of life. Passeriformes (perching birds) represent the largest and most diverse order of birds, comprised of over 6,500 species, more than half of all living birds. This clade originated approximately 50-60 million years ago in Australia and rapidly spread globally, occupying nearly all terrestrial habitats. Passeriformes exhibit remarkable diversity in ecological strategies, including varied diets and diverse breeding systems. Migratory behavior varies considerably across the clade, with some species being long-distance migrants while others remain sedentary residents. Chromosome number is a fundamental genomic trait that can influence speciation rates and genome evolution. Previous comparative studies in Passeriformes have examined traits such as song learning, brain size, plumage coloration, and morphology, but the relationship between chromosome number and migratory status remains unexplored.

Research question: Is there an association between chromosome number and migratory status in passerine birds, and does this relationship persist after accounting for shared evolutionary history?

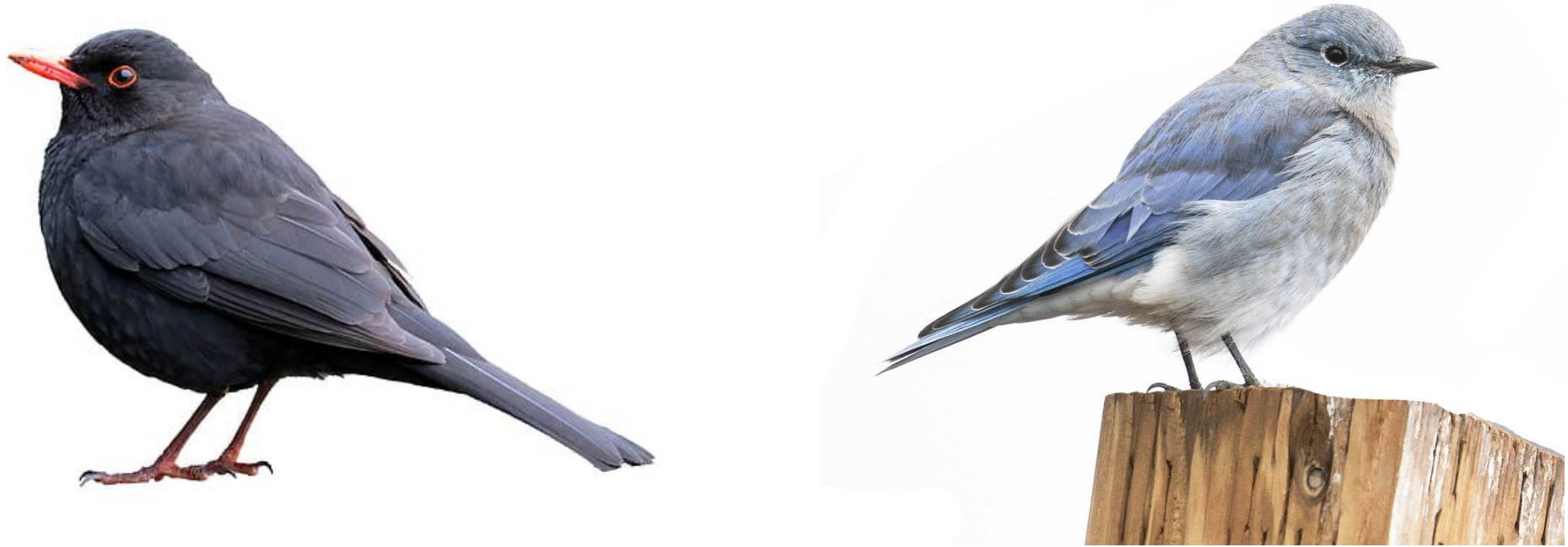
Study Design & AI

- Trait Selection:
- Two traits were selected for analysis based on data availability and biological interest
    - Chromosome number (continuous trait): obtained from Dr. Blackmon
    - Migratory status (categorical: migratory vs. non-migratory): obtained from ornithological databases
  - Data Sources:
    - Chromosome data:
      - title = {From Micro to Macro:Avian Chromosome Evolution is Dominated by Natural Selection}, author = {Alfieri H., Bolwerk K., Hu Z., Blackmon H.}, journal = {bioRxiv}, year = {2024}, doi = {10.1101}
    - Migration data
      - BirdLife. "BirdLife Data Zone." [Datazone.birdlife.org](https://datazone.birdlife.org/), 2024, [datazone.birdlife.org/](https://datazone.birdlife.org/)
      - Birds of the World, Cornell Lab, <https://birdsoftheworld.org/bow/home>
      - Animalia, Evgen, Ivan, Ira, <https://animalia.bio/>
    - Phylogeny
      - Species-level passerine phylogeny (Prasetya et al. 2025)

AI Assistance:  
AI tools (Claude) were used to develop R scripts for phylogenetic data processing and visualization, generate code for statistical analyses, and troubleshoot coding errors.

Model Fitting

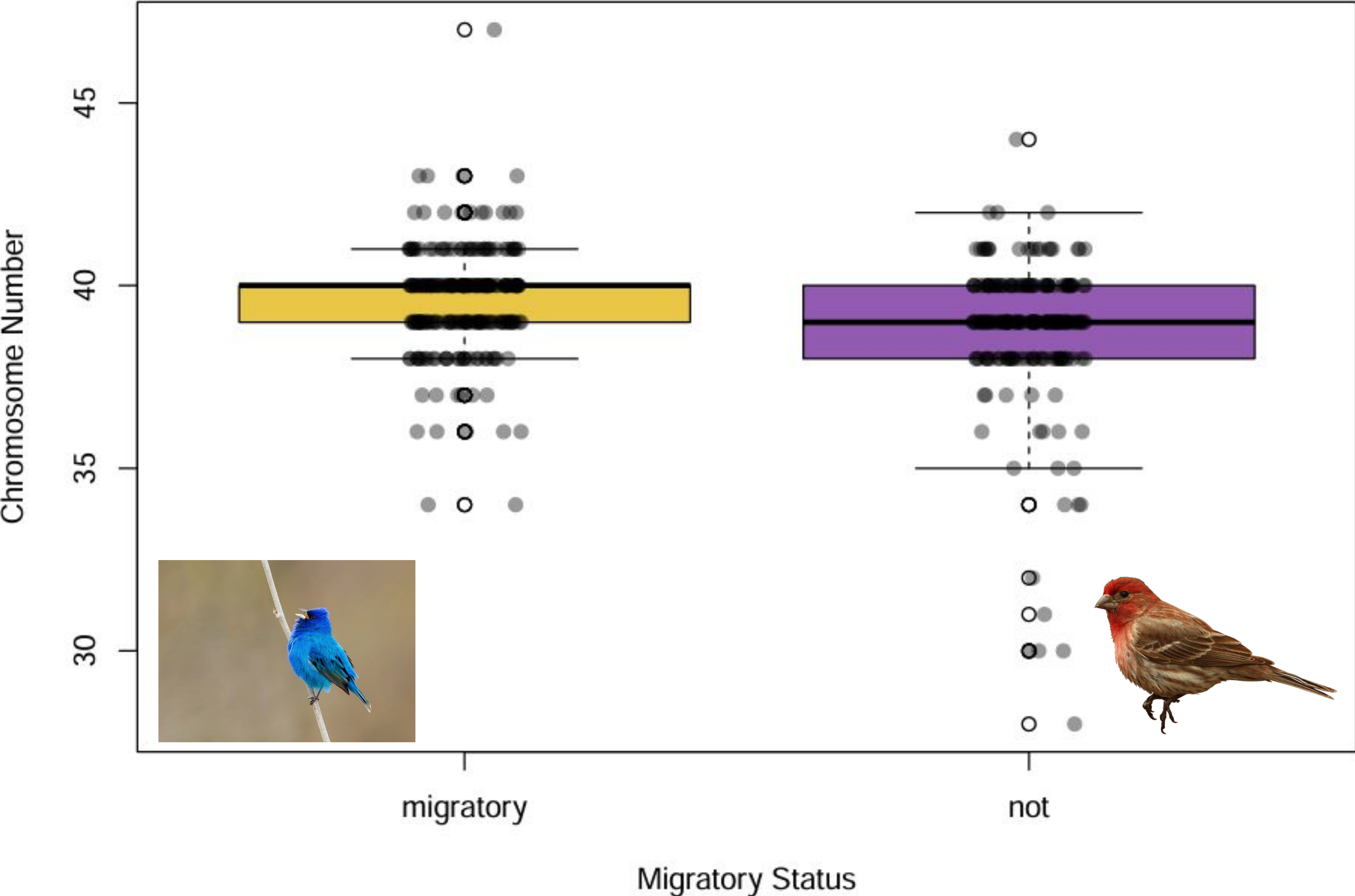
Since species share evolutionary history, and aren't statistically independent, we employed phylogenetic comparative methods implemented in R packages "ape" and "phytools". Two-sample t-test and Wilcoxon rank-sum test gave a comparison of group means and non-parametric comparison, respectively, but neither account for phylogeny. Phylogenetic ANOVA compared the groups while controlling for relatedness, and Blomberg's K measured phylogenetic signal in traits while still accounting for phylogeny. The Phylogenetic ANOVA simulated null distribution using 1,000 permutations, and tests whether trait differences between groups exceed expectation given phylogenetic structure. Phylogenetic signal using Blomberg's K evaluated the signal. A K value of zero indicated no signal, less than 1 indicated less signal than expected under Brownian motion, a value of 1 indicated evolution consistent with Brownian motion, and greater than 1 showed strong phylogenetic conservatism.



Results

The mean chromosome number for migratory (39.59) species and non-migratory (38.79) species had a difference of 0.80. The T-test resulted in  $t=4.58$ ,  $df=360.2$ , with a p-value of  $<0.0001$  and Wilcoxon had a p-value  $<0.0001$ . Phylogenetic ANOVA had an F-value=21.86 and p-value=0.027. Phylogenetic signal gave chromosome number  $K=0.38$ , p-value=0.001, and migratory status  $K=0.28$ , p-value=0.001. These findings suggest migratory passeriforme species have significantly (statistically) higher chromosome numbers than non-migratory species. This association remains significant ( $p=0.027$ ) even after accounting for phylogenetic relatedness using phylogenetic ANOVA.

Chromosome Number by Migratory Status



Discussion

The phylogenetic ANOVA revealed a significant difference in chromosome number between migratory and non-migratory species ( $F=21.86$ ,  $p=0.027$ ), indicating that migratory status is associated with chromosome number variation, even after controlling for shared ancestry. The probability of observing this difference by random chance is only 2.7%. The t-test p-value ( $<0.0001$ ) is much smaller than the phylogenetic ANOVA p-value (0.027) because the t-test treats all species as independent observations. Closely related species tend to share similar traits due to common ancestry, artificially inflating sample size and statistical significance. The phylogenetic ANOVA corrects for this non-independence, providing a more conservative and accurate test. Both traits show significant but moderate phylogenetic signal ( $K<1$ ), indicating that while related species tend to be similar, both chromosome number and migratory behavior have evolved multiple times across the passeriforme phylogeny. This evolutionary lability allows us to detect correlated patterns between traits. Several mechanisms could explain the association between higher chromosome numbers and migratory behavior. One reason could be chromosomal rearrangements, migration-related genes may be associated with genomic regions prone to fission events. Another reason could be selective pressures, there could be different genome structural constraints between migratory and sedentary lifestyles. The other reason could be due to historical fission events, migratory lineages may have experienced chromosome fission during their evolutionary history. Some limitations to these results are that correlation doesn't imply causation, the difference in chromosome number (0.80) is relatively small, and there are likely unmeasured confounding variables that may influence both traits.

Conclusions

This study demonstrates a significant association between chromosome number and migratory status in passeriforme birds. Migratory species possess, on average, 0.80 more chromosomes than non-migratory species. Critically, this relationship persists after phylogenetic correction ( $p=0.027$ ), indicating it's not solely due to shared evolutionary history. Both traits exhibit moderate phylogenetic signal, suggesting evolutionary conservation alongside repeated independent evolution across the passeriforme tree. These findings contribute to our understanding of genome evolution in relation to complex behavioral traits and highlight the importance of using phylogenetic comparative methods when analyzing species-level data.